

1. Cup Product

1.1. Problem 13

Describe $H^*(\mathbb{C}P^\infty/\mathbb{C}P^1; \mathbb{Z})$ as a ring with finitely many multiplicative generators. How does this ring compare with $H^*(S^6 \times \mathbb{H}P^\infty; \mathbb{Z})$

Proof All coefficients are $\mathbb{Z}$ coefficients unless otherwise stated. $(\mathbb{C}P^\infty, \mathbb{C}P^1)$ is a good pair, thus

$$H^n(\mathbb{C}P^\infty/\mathbb{C}P^1) = H^n(\mathbb{C}P^\infty, \mathbb{C}P^1)$$

Note $\mathbb{C}P^\infty/\mathbb{C}P^1$ has a 0 cell and $2n$ cells for all $n \geq 2$. Thus $H^n(\mathbb{C}P^\infty/\mathbb{C}P^1) = 0$ for $n = 1, 2$. By the cohomological long exact sequence of pairs, $H^n(\mathbb{C}P^\infty, \mathbb{C}P^1) \cong H^n(\mathbb{C}P^\infty)$. Let $\alpha \in H^4(\mathbb{C}P^\infty/\mathbb{C}P^1)$ and $\beta \in H^6(\mathbb{C}P^\infty/\mathbb{C}P^1)$ be generators where $q^*(\alpha) = x^2$ and $q^*(\beta) = x^3$ for x being the generator of $H^*(\mathbb{C}P^\infty)$. Let $q : \mathbb{C}P^\infty \rightarrow \mathbb{C}P^\infty/\mathbb{C}P^1$ be the standard quotient map.

$$q^*(\alpha^3 - \beta^2) = x^6 - x^6 = 0$$

Since q is a isomorphism, $\alpha^3 - \beta^2 = 0 \Rightarrow \alpha^3 = \beta^2$. Thus,

$$H^*\left(\frac{\mathbb{C}P^\infty}{\mathbb{C}P^1}\right) = \frac{\mathbb{Z}[\alpha, \beta]}{(\alpha^3 - \beta^2)}$$

where $|\alpha| = 4$ and $|\beta| = 6$.

$$H^*(S^6 \times \mathbb{H}P^\infty) = \frac{\mathbb{Z}[a, b]}{(b^2)}$$

Where $|a| = 4$ and $|b| = 6$. Both rings are not isomorphic. □

1.2. Problem 14

Let $q : \mathbb{R}P^\infty \rightarrow \mathbb{C}P^\infty$ be the natural quotient map by regarding both spaces as quotients of S^∞ , modulo multiplication of real scalars in one case and complex scalars in the other.

Show that $q^* : H^*(\mathbb{C}P^\infty; \mathbb{Z}) \rightarrow H^*(\mathbb{R}P^\infty; \mathbb{Z})$ is surjective in even dimensions by showing first a geometric argument that the restriction $q : \mathbb{R}P^2 \rightarrow \mathbb{C}P^1$ induces a surjection on H^2 and then appealing to the cup product structures. Next, form a quotient space $X = \mathbb{R}P^\infty \sqcup \mathbb{C}P^\infty$ by identifying each point $x \in \mathbb{R}P^{2n}$ with $q(x) \in \mathbb{C}P^n$. Show there are ring isomorphisms $H^*(X; \mathbb{Z}) \cong \frac{\mathbb{Z}[\alpha]}{(2\alpha^{n+1})}$ and $H^*(X; \mathbb{Z}_2) \cong \mathbb{Z}_2[\alpha, \beta]/(\beta^2 - \alpha^{2n+1})$, where $|\alpha| = 2$ and $|\beta| = 2n + 1$.

Proof □

1.3. Problem 15

For a fixed coefficient field F , define a Poincaré Series of a space X to be the formal power series $p(t) = \sum_i a_i t^i$ where $a_i = \dim H^i(X; F)$ as a vector space over F , assuming $\dim$ is finite for all i . Compute the power series for $S^n, \mathbb{R}P^n, \mathbb{R}P^\infty, \mathbb{C}P^n, \mathbb{C}P^\infty$

1. $S^n: p(t) = 1 + t^n$
2. $\mathbb{R}P^n$:
 - If F has characteristic 2, then $p(t) = \sum_{i=0}^{n+1} t^i = \frac{1-t^{n+1}}{1-t}$
 - If $\text{char}(F) \neq 2 \Rightarrow H^i = F \Leftrightarrow i \bmod 2 = 1$ or $i = 0$.
3. $\mathbb{R}P^\infty$

Prove that $p(X)p(Y) = p(X \times Y)$

Proof By the Kunneth formula $H^*(X; F) \otimes H^*(Y; F) = H^*(X \times Y; F)$. Thus

$$H^n(X \times Y; F) = \bigoplus_{p+q=n} H^p(X; F) \otimes H^q(Y; F)$$

Thus

$$\begin{aligned} \dim H^n(X \times Y; F) &= \dim \left(\bigoplus_{p+q=n} H^p(X; F) \otimes H^q(Y; F) \right) \\ &= \sum_{p=0}^n (\dim(H^p(X; F))) (\dim(H^{n-p}(Y; F))) \end{aligned}$$

Thus

$$p(X \times Y)(t) = \sum_{n \geq 0} \left(\sum_{p=0}^n (\dim(H^p(X; F))) (\dim(H^{n-p}(Y; F))) \right) t^n = p(X)(t)p(Y)(t)$$

□

1.4. Problem 16

Show that if X and Y are finite CW complexes such that $H^*(X; \mathbb{Z})$ and $H^*(Y; \mathbb{Z})$ contain no elements of order a power of a given prime p , then the same is true for $X \times Y$

Proof Unless otherwise specified, $\mathbb{Z}$ coefficients are used. By the Cohomological UCT,

$$H^n(X) = \text{Ext}(H_{n-1}(X), \mathbb{Z}) \oplus \text{hom}_{\mathbb{Z}}(H_n(X), \mathbb{Z})$$

Thus no elements of order some power of p in $H^n(X) \Rightarrow$ no elements of some power of p in $\text{Ext}(H_{n-1}(X), \mathbb{Z})$ or $\text{hom}_{\mathbb{Z}}(H_n(X), \mathbb{Z})$.

$\text{Ext}(H_{n-1}(X), \mathbb{Z})$ has no elements of some power of p : Suppose $\exists m \in \mathbb{Z} \ni \exists g \in H_{n-1}(X) \ni |\langle g \rangle| = p^m$. WLOG, assume m is maximal. Thus Since X is a finite CW complex, $H_{n-1}(X)$ is a finitely generated abelian group. Thus $H_{n-1}(X) = \mathbb{Z}^r \oplus \mathbb{Z}_{p^{i_1}} \oplus \dots$. Thus $\text{Ext}(H_{n-1}(X), \mathbb{Z}) = \mathbb{Z}_{p^{i_1}} \oplus \dots$, thus $\text{Ext}(H_{n-1}(X), \mathbb{Z})$ has elements of order p^m . Thus by contrapositive, $\text{Ext}(H_{n-1}(X), \mathbb{Z})$ has no elements of some power of p then $H_{n-1}(X)$ has no elements of some power of p for all n .

Thus $H_{n-1}(X)$ and $H_n(X)$ hve no elements of order p^m for all m . By the Universal coefficient theorem for Homology, $H_n(X; \mathbb{Z}_p) = H_n(X) \otimes \mathbb{Z}_p \oplus \text{Tor}(H_{n-1}(X), \mathbb{Z}_p) = \mathbb{Z}_p^{\text{rank}(H_n(X))}$. Proof is identical for Y .

$$\begin{aligned} \dim(H^n(X \times Y; \mathbb{Z}_p)) &= \sum_{p=0}^n \dim(H^p(X; \mathbb{Z}_p) \otimes H^{n-p}(Y; \mathbb{Z}_p)) \\ &= \sum_{p=0}^n \dim(H_p(X; \mathbb{Z}_p)) \dim(H_{n-p}(Y; \mathbb{Z}_p)) \\ &= \sum_{p=0}^n \text{rank}(H^p(X)) \text{rank}(H^{n-p}(Y)) \\ &= \text{rank}(H^n(X \times Y)) \end{aligned}$$

This is true $\Leftrightarrow H^*(X \times Y)$ has no factors of order p^m for all $m \in \mathbb{N}$.

□

1.5. Problem 18

Theorem 1.5.1 For the closed orientable surface M for genus $g \geq 1$, for each nonzero $a \in H^1(M; \mathbb{Z})$, $\exists b \in H^1(M; \mathbb{Z}) \ni \alpha\beta \neq 0$. Thus M is not homotopy equivalent to a wedge sum $X \vee Y$ of CW complexes with nontrivial reduced homology

Proof

$$H^*(M; \mathbb{Z}) = \frac{\mathbb{Z}[\alpha_1, \beta_1, \dots, \alpha_{2g}, \beta_{2g}]}{(\alpha_i \alpha_j, \beta_i \beta_j, \alpha_i \beta_j, \alpha_i^2, \beta_i^2, \alpha_i \beta_i + \beta_i \alpha_i, \beta_i \alpha_i \beta_i, \alpha_i \beta_i \alpha_i)}$$

Let $a \in H^1(M; \mathbb{Z}) \ni a \neq 0$, $a = \sum_{i=1}^{2g} (c_i \alpha_i + d_i \beta_i)$. There exists some $i \in [1, \dots, 2g] \ni \alpha_i$ or $\beta_i \neq 0$. WLOG $i = 1$ and $\alpha_i \neq 0$ (Argument is symmetric for β_i). $a\beta_i = c_i(\alpha_i \beta_i)$

Assume for contradiction, that $M = X \vee Y$ where $\tilde{H}_1(X) \neq 0 \neq \tilde{H}_1(Y) \Rightarrow H_1(X) \neq 0 \neq H_1(Y)$. Thus $H^1(M) = H^1(X) \oplus H^1(Y)$. Now take any two basis elements in X and Y and now they cannot multiply to nonzero number. □